

Phase I/II Operating Characteristics

Old versus new allocation design - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
One-stage - Old	10.1	8.1	4.3	0.9	0.1	76.5
One-stage - New	17	7.2	1.7	0.1	0.1	73.9
Two-stage (highest utility) - Old	10	9.8	3.9	0.8	0	75.5
Two-stage (highest utility) - New	18	7.2	1.2	0	0.2	73.4
Two-stage (top-2 randomized) - Old	12.1	7.3	2.8	1.1	0	76.7
Two-stage (top-2 randomized) - New	17.4	7.2	2	0.8	0.1	72.5
Mean patients treated						
One-stage - Old	11.7	4.3	1.7	0.7	0.2	
One-stage - New	11.1	4.4	2.3	0.9	0.3	
Two-stage (highest utility) - Old	11.6	4.4	1.8	0.7	0.2	
Two-stage (highest utility) - New	11.1	4.4	2.3	1	0.3	
Two-stage (top-2 randomized) - Old	11.3	4.6	1.9	0.7	0.2	
Two-stage (top-2 randomized) - New	11	4.4	2.3	1	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
One-stage - Old	28.6	23.4	19.6	17.6	10.8	0
One-stage - New	29.3	21.3	20.2	18.8	10.4	0
Two-stage (highest utility) - Old	30.5	23.7	18	15.9	11.9	0
Two-stage (highest utility) - New	34.1	21.3	16.6	16.4	11.5	0.1
Two-stage (top-2 randomized) - Old	33	27.3	17.2	13.2	9.2	0.1
Two-stage (top-2 randomized) - New	33.9	22.2	16.5	15.4	12	0
Mean patients treated						
One-stage - Old	8	6.8	5.7	5.3	4.2	
One-stage - New	7.3	6.7	5.7	5.7	4.6	
Two-stage (highest utility) - Old	7.7	6.8	5.7	5.4	4.4	
Two-stage (highest utility) - New	7.6	6.8	5.7	5.5	4.4	
Two-stage (top-2 randomized) - Old	7.7	6.8	5.6	5.5	4.4	
Two-stage (top-2 randomized) - New	6.6	6.1	5.6	6.1	5.6	

Source: paired old/new 1,000-trial dose summaries; shared-alpha additive model, continuous enrollment, IPDE alpha = 0.

Utility 3 scores: no toxicity/no efficacy = 40; no toxicity/efficacy = 100; toxicity/no efficacy = 0; toxicity/efficacy = 60. Stop % is no-OBD selection.

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Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
One-stage - Old	10.2	49.6	23.3	10.9	2.6	3.4
One-stage - New	14.4	43.9	25.7	10.8	2.4	2.8
Two-stage (highest utility) - Old	9.8	48.7	25.2	11.3	2.2	2.8
Two-stage (highest utility) - New	14.2	47.4	25.8	7.7	1.7	3.2
Two-stage (top-2 randomized) - Old	7.1	53.2	22.7	11	3.4	2.6
Two-stage (top-2 randomized) - New	10.4	47.7	25.6	10.6	2.7	3
Mean patients treated						
One-stage - Old	6.3	11	6.7	4.1	1.4	
One-stage - New	5.2	11	7.6	4.2	1.4	
Two-stage (highest utility) - Old	6	10.2	7.2	4.5	1.7	
Two-stage (highest utility) - New	5.3	11.2	7.9	3.8	1.3	
Two-stage (top-2 randomized) - Old	7	9.5	7.2	4.3	1.6	
Two-stage (top-2 randomized) - New	5.7	9.5	8	4.6	1.8	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
One-stage - Old	1.4	1.5	5.2	58.8	33.1	0
One-stage - New	1.4	0.9	5	58.6	34.1	0
Two-stage (highest utility) - Old	1.5	1.6	6	55.7	35.2	0
Two-stage (highest utility) - New	2	1.5	6.2	55.5	34.8	0
Two-stage (top-2 randomized) - Old	0.8	1	4.4	56.9	36.9	0
Two-stage (top-2 randomized) - New	0.4	0.8	3.6	56.6	38.6	0
Mean patients treated						
One-stage - Old	3.5	3.5	4.4	12.3	6.3	
One-stage - New	3.4	3.5	4.4	12.2	6.5	
Two-stage (highest utility) - Old	3.5	3.5	4.2	10.7	8.2	
Two-stage (highest utility) - New	3.4	3.5	4.4	10.6	8.1	
Two-stage (top-2 randomized) - Old	3.9	4	4.9	9.2	8	
Two-stage (top-2 randomized) - New	3.6	3.7	4.9	9.4	8.4	

Source: paired old/new 1,000-trial dose summaries; shared-alpha additive model, continuous enrollment, IPDE alpha = 0.

Utility 3 scores: no toxicity/no efficacy = 40; no toxicity/efficacy = 100; toxicity/no efficacy = 0; toxicity/efficacy = 60. Stop % is no-OBD selection.

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
One-stage - Old	14.1	24	44	11	0.7	6.2
One-stage - New	16.4	22.3	43.7	10.9	1.1	5.6
Two-stage (highest utility) - Old	15.1	22.4	44.8	10.6	1.3	5.8
Two-stage (highest utility) - New	16.3	23.9	44	8.4	0.9	6.5
Two-stage (top-2 randomized) - Old	11.6	24.2	44.8	12.7	1.1	5.6
Two-stage (top-2 randomized) - New	10.5	23.6	46.6	12.5	1.1	5.7
Mean patients treated						
One-stage - Old	7.9	7.5	9.4	3.6	0.6	
One-stage - New	7.1	7.9	9.8	3.6	0.7	
Two-stage (highest utility) - Old	7.4	7.4	9.3	4	1.1	
Two-stage (highest utility) - New	7.7	7.9	9.4	3.2	0.9	
Two-stage (top-2 randomized) - Old	8.3	8.1	8	3.9	1.1	
Two-stage (top-2 randomized) - New	7.4	8.2	8.4	4	1.1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
Utility 3	41	40.8	42.4	43	58	
OBD selection (%)						
One-stage - Old	8.9	10.9	9.1	9.5	60.4	1.2
One-stage - New	14	7.9	6	7.8	63.2	1.1
Two-stage (highest utility) - Old	10.1	11	8.9	9	60.1	0.9
Two-stage (highest utility) - New	17.9	10	7.3	7.5	56	1.3
Two-stage (top-2 randomized) - Old	9.2	10.8	9.3	8.9	60.8	1
Two-stage (top-2 randomized) - New	11.7	6	5.5	8.5	67.3	1
Mean patients treated						
One-stage - Old	5.6	5	4.5	4.8	9.9	
One-stage - New	4.5	4.5	5	5.3	10.4	
Two-stage (highest utility) - Old	5.4	5	4.5	4.9	10	
Two-stage (highest utility) - New	4.8	4.7	5.3	5.4	9.6	
Two-stage (top-2 randomized) - Old	6.5	6	5	5	7.4	
Two-stage (top-2 randomized) - New	4.7	4.8	5.3	6.9	8.2	

Source: paired old/new 1,000-trial dose summaries; shared-alpha additive model, continuous enrollment, IPDE alpha = 0.

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